EL NIÑO ACTION BRIEFING · 14 OCTOBER 2026

South (CASUD), Réunion

Not an El Niño drought: a record carried-over water deficit, leaking networks, heat, mosquito-borne disease bred in stored water, leptospirosis, and cyclones that become water outages, in a hot El Niño year. In the south, the water is already rationed, the networks lose more than four litres in ten, and the people who die are the ones nobody has listed.

IMPACT REGION	EXCESS HEAT DEATHS	WINDOW	DOMINANT HAZARD
All Réunion (one region); CASUD shares the Saint-Pierre arrondissement row	About 0.8 by population share of the arrondissement's 2 (our arithmetic; Réunion: 5)	6-month, Sep 2026 – Feb 2027	Rationed, leaking water; leptospirosis; people outside the system in a cyclone

17/17 recommendations checked. 1 confirmed, 8 corrected, 0 unverified; 8 contraindicated (the Do not cards). **The funding target fell 38%**, from the panel's €802,261 to €496,651: the panel put €607,410 into water kits for every poor household in Saint-Joseph, which one CCAS cannot deliver before the peak, and the outreach line was cut back to what the State's shelter budget does not already pay. Two inherited figures were corrected: the South's leptospirosis "39 per 100,000" is 39% of cases (about 18 per 100,000), and the Rivière des Remparts dyke was finished in 2022. Exposure: Climate Impact Lab, *451,000 at Risk*, Sep 2026, pre-peer-review, heat only. **The CASUD figure is our scaling, not an Impact Lab finding.**

EXECUTIVE SUMMARY — AT A GLANCE



21.30°S 55.60°E, France · IGN ADMIN EXPRESS, OpenStreetMap contributors; Natural Earth

- **The risk.** Belal's one death here (2024) was a homeless man in a tin shack in Le Tampon; on 9 October 2026 Le Tampon went three days without safe tap water.
- **First action.** By 30 November, the CCAS of Le Tampon and Saint-Joseph list everyone sleeping rough by name, so a known face asks again at each orange alert.
- **The money.** €295k / €497k / €904k (floor / target / ceiling). Stocking Saint-Philippe for a cut-off comes first after the outreach (€135k).
- **Do not.** Never answer Saint-Joseph's groundwater *crise* by pumping coastal wells harder: seawater intrusion there is irreversible.

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* Suggested formulation: check with a health worker before sharing.

CASUD (Communauté d'agglomération du Sud) is Le Tampon, Saint-Joseph, Saint-Philippe, and Entre-Deux: 134,044 people (INSEE, populations légales 2023, in force 2026) in 55,979 households. CASUD holds the drinking-water competence and runs it through one contract for all four communes: a delegation to **Sudéau** (a SAUR group company) from 1 July 2023 to 30 June 2033, serving 61,639 subscribers on 1,133 km of pipe. The network delivered **57.9%** of the water it produced in 2024 (59.7% in 2023), below the island's 63%. Saint-Philippe is the end of the line: one coastal road, the RN2, which lava cut to the east in March 2026 and floods cut to the west in August 2021. This briefing is the fourth of six for Réunion. It is written to be adopted as an annex to CASUD's intercommunal safeguard plan (PICS), *volet pénurie d'eau, chaleur et épidémies*, and each action carries its cyclone phase (avant, pendant, or sauvegarde).

COMMUNE	HOUSEHOLDS (2023)	POVERTY (FILOSOFI 2023)	VIGIEAU, DRINKING WATER (14 OCT 2026)	NETWORK EFFICIENCY (2021)
Le Tampon	34,653	38.1%	Alerte	72% (Grenelle target 70%)
Saint-Joseph	16,240	43.8%	**Crise** (groundwater crise; surface water alerte renforcée)	50% (target 75%)
Saint-Philippe	2,156	43.0%	Alerte renforcée	54% (target 69%)
Entre-Deux	2,930	31.0%	Alerte	67% (target 69%)

VigiEau shows the restriction level, not the state of supply; prefectoral order of 9 October 2026, in force 10 October, read on 14 October. Commune efficiency: Office de l'eau, *Chroniques de l'eau* n°133 (2023), 2021 data; service-wide 2024 figure: CASUD's annual water report (RPQS 2024). Poverty is a rate per person, so applying it to households (about 22,150 poor households) is our approximation. Seven priority neighbourhoods (QPV): La Châtoire, Les Trois Mares, Les Araucarias, and Centre-ville in Le Tampon; Cayenne-Butor-Les Quais, Centre Ville-Cités, and Langevin in Saint-Joseph; none in Saint-Philippe or Entre-Deux. Supply is mixed: groundwater was 41.8% of CASUD's abstraction in 2024. El Niño on Réunion: rain on average in surplus, a clearly hotter season, and no change in the island's cyclone exposure (Météo-France). The deficit comes from before it: 2025–26 was the driest rainy season in 54 years of records (−45%, provisional; −60 to −70% in the hauts), the second deficit season in a row after −30% in 2024–25 (Météo-France, provisional season review, 29 April 2026). **Supply is already failing:** on 9

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October 2026 a breakdown at Le Tampon's Leveneur plant cut water from the 14th km to the town centre, and a do-not-drink order followed; residents went three days without safe tap water, and the commune handed out water packs.

WHAT THIS PLACE SHOULD DO RANKED

Ranked by expected lives saved per unit of effort before the peak. Backing lenses are shown per action; an action several lenses reached independently is the stronger bet.

1. By 30 November, list by name everyone sleeping rough, in tin shacks, or in ravine camps in Le Tampon and Saint-Joseph, the Hauts included, so that a known face offers shelter before each orange alert and comes back on the day

PARTIALLY CONFIRMED | PHASE: AVANT | EMERGENCY MANAGER · FRONTLINE · HISTORIAN · SKEPTIC · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
CCAS of Le Tampon and Saint-Joseph, with the Croix-Rouge (Le Tampon antenne, Samu social) and the Saint-Joseph Boutique Solidarité	List by 30 November 2026; rounds before each orange alert	€22,800: kits, transport, list costs (F01)	Cyclone deaths among people who refuse or never hear the shelter offer

In Belal (15 January 2024), all three of Réunion's dead were homeless. One was in CASUD: a man found in a *cabane en tôles* in Le Tampon the day after the storm. Réunion La 1ère reported that the Saint-Gilles man had refused shelter; the Interior Minister said others 'auraient refusé' (conditional). Where an offer was made, it failed at the last metre. The Croix-Rouge's Samu social runs six mobile teams across the island, but not in the Hauts (Fondation Abbé Pierre, 2024), and the Hauts are where much of Le Tampon lives. No count of rough sleepers exists for CASUD; Saint-Pierre, next door, counted 57 in 2023. The list is made in November by the people who already know them: the Croix-Rouge day centre in Le Tampon, the Saint-Joseph Boutique Solidarité, the CCAS, shopkeepers, and parishes. Each person gets a named worker who asks twice: once in November, and again before the orange alert, with transport to the shelter.

Corrections: there was one Belal death in Le Tampon, not two (a 24matins report is wrong; the others were at Saint-Gilles and in Salazie). No cause of death was published. The minister's 'auraient refusé' is conditional, not an established fact for each person. The panel's €60,000 for a salaried post was cut: maraudes are normally paid from the State's emergency-shelter budget, so donor money buys the list,

* Suggested formulation: check with a health worker before sharing.

the kits, and the transport, not a post.

2. From November, the ARS writes to every GP in CASUD to treat suspected leptospirosis without waiting for a test, and takes the free vaccine and boots to farmers and to CASUD and commune crews *

PARTIALLY CONFIRMED | PHASE: AVANT | PUBLIC HEALTH · SKEPTIC · FRONTLINE · EMERGENCY MANAGER | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
ARS La Réunion, with the CGSS; occupational health services for CASUD, Sudéau, and commune crews	GP letter by 15 November; first vaccine doses in November	Within ARS and CGSS budgets (the vaccine is free)	Leptospirosis deaths from late treatment after the rains return

The South had more leptospirosis cases than any other sector from 1 January to 15 April 2026: 56, or 39% of the island's total. **Le Tampon (20) tied with Saint-Benoît, in the East (20), for the most cases of any commune, and Saint-Joseph had 15**, in the driest rainy season in 54 years of records (SPF, 17 April 2026). Of 100 cases investigated, 69% had done farm or garden work, and 19% had used untreated water for watering, irrigation, or washing. Réunion had 250 cases and 3 deaths this year to 9 October. Delay is what kills: on Réunion, the median delay before antibiotics was 4 days, and patients first seen by a GP rather than in an emergency unit were far more likely to be treated late (OR 4.63, 1.43–14.93; Lebeau-Desmoulin and others, 117 hospital patients, 2014–15). So the letter tells GPs to treat a fever after mud, water, or farm contact on suspicion. The vaccine has been free for all adults since 29 September 2025 (ARS and CGSS), with the most exposed prioritised. It is given at the CHU Nord, the CHU Sud, and occupational health services: crews get it through theirs, and reaching farmers would need the ARS to run mobile sessions.

* Suggested formulation: check with a health worker before sharing.

Corrections: the set's "South 39 per 100,000" (used in the cirques briefing) is a misreading: 39 is the South's percentage of cases. The East had the highest rate (34 per 100,000); the South's is about 18. The vaccine (Spirolept) has never had a clinical efficacy trial and covers one serogroup (Icterohaemorrhagiae) only; it takes two injections 15 days apart, a booster at 4–6 months, then one every 2 years if exposure continues. Protection builds over the first month after two doses; it is not this season's main defence (RE-X08). It sits alongside prevention, not ahead of it: boots, gloves, covered wounds, no wading, and early treatment are the main protection.

3. Before 15 December, stock Saint-Philippe for seven days: a water-storage kit for each poor household and a three-day food kit for every household, held by the CCAS and released at pré-alerte or alerte orange, before the RN2 is cut

PARTIALLY CONFIRMED | PHASE: PENDANT | HISTORIAN · EMERGENCY MANAGER · FOOD SUPPLY · PHILANTHROPIST · SKEPTIC | ADAPTATION INVENTORY: ANTICIPATORY CASH (BANGLADESH, NEPAL), NOT STOCK

OWNER	BY WHEN	COST	PREVENTS
Commune and CCAS of Saint-Philippe, with the Croix-Rouge / PIROI (water kits) and the Banque alimentaire des Mascareignes (food)	Ask PIROI by 1 November; stock in place by 15 December 2026	€134,851 (F02)	Households without safe water or food while the only road is cut

Saint-Philippe has 2,156 households and a 43% poverty rate, at the end of one road. The road fails from both sides. To the east, lava crossed the RN2 at the Grand Brûlé on 13 March 2026, for the first time since 2007, and crossed again during the February–April eruption; the road reopened fully only at the end of May, and in the meantime the trip to Sainte-Rose went round the island and took 2.5 hours instead of 30 minutes (RCI, 13 March 2026). To the west, on 28 August 2021, floods cut the RN2 at Vincendo and isolated Grand Galet and Langevin. After Garance (2025), EDF had to re-power the Australine borehole at Saint-Philippe: here, a power cut is a water cut. The arithmetic: 927 poor households × €85 water kit = €78,795, plus 2,156 households × €26 food kit = €56,056, so €134,851. Ask PIROI first what it already holds on the island, and buy only the difference.

Corrections: lava at the Grand Brûlé cuts Saint-Philippe off from the east only; supply from Saint-Joseph still gets through. The cut that isolates the commune is a flood or slide on the western side, or an alerte rouge, when all traffic stops. So the kits go out at pré-alerte or alerte orange, before the cut: at alerte rouge everyone is confined and nothing can be handed out. The "30 minutes to 2.5 hours" detour is from RCI, not from the parliamentary question it was first attributed to (QE 17-14724). No record was found of Saint-Philippe being

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isolated in Belal or Garance. The Bangladesh anticipatory-cash study (Pople and others, WFP, 2021) concerns cash, not stock, and is cited only for "early beats late". Neither the Saint-Philippe CCAS nor the Banque alimentaire (based at La Saline) has a record of moving stock at this scale here; that is the largest risk to the line.

4. Before the season, the AURAR dialysis units and the CHU Sud bring their patient lists and early-session plans up to date, the four CCAS run their vulnerable-people registers, and at pré-alerte dialysis patients get their session early, with a transport plan for Saint-Philippe

PARTIALLY CONFIRMED | PHASE: AVANT | PUBLIC HEALTH · HISTORIAN · EMERGENCY MANAGER | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
ARS La Réunion, with AURAR (Saint-Joseph and Le Tampon units), the CHU Sud, and the four CCAS	Plans by 30 November 2026; early sessions at each pré-alerte	Staff time; sessions are reimbursed	Deaths from missed dialysis and from heat or water cuts among people living alone

Réunion needs about five times as much dialysis as mainland France, with end-stage kidney failure three to four times as common (Senate report r25-554, April 2026). AURAR runs assisted self-dialysis units in Saint-Joseph and Le Tampon; there is none in Saint-Philippe, so its patients travel the RN2. After Hurricane Sandy, haemodialysis patients dialysed early had lower odds of death at 30 days (adjusted OR 0.72, 0.52–0.997) and of an emergency-department visit (OR 0.80) (Lurie and others, *AJKD* 2015, Medicare patients in New York and New Jersey). The CCAS registers (CASF L121-6-1) let the commune phone people aged 65 and over and adults with disabilities on hot days, at an alert, and on day 2 of a water cut. Réunion has no heat vigilance, so the CCAS sets its own trigger.

* Suggested formulation: check with a health worker before sharing.

Corrections: the register is voluntary: people ask to be on it, so a CCAS cannot hold a list of "everyone", and an ARS-held list across providers runs into medical secrecy. Each provider checks its own patients instead. Early sessions work for a cyclone, which is forecast; a lava cut is not, so for that case the plan is transport. The Sandy study is observational and from a US city; the mechanism (sessions missed when roads are cut) carries over.

5. In Saint-Joseph, at *crise* now: every cut or restriction notice names where to get safe drinking water and warns against ravine water, and the poorest households get a water-storage kit

PARTIALLY CONFIRMED | PHASE: AVANT | SKEPTIC · EMERGENCY MANAGER · PHILANTHROPIST · FRONTLINE | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
CASUD, with Sudéau, the commune and CCAS of Saint-Joseph, and the ARS	Notice wording now; kits by 15 December 2026	€170,000 for 2,000 kits (F03); notices at no cost	Drinking or washing in untreated water; dengue bred in uncovered storage

Saint-Joseph is the only commune on Réunion at drought *crise* for drinking water. Its network delivered half the water it produced in 2021 (50%, against a target of 75%). A notice that lists cut hours and says nothing else leaves people to find water themselves, and in this commune that means ravines and basins: 19% of this year's investigated leptospirosis cases had used untreated water. So every CASUD and Sudéau notice says, in French and Creole, where the bottled or approved drinking water is, and carries one line: "ravine or basin water: not for drinking or washing, not on wounds; boots and gloves to irrigate". It also carries the island rule RE-01: "couvrez vos réserves". The CCAS gives the 2,000 poorest households a water-storage kit (two covered 20-litre jerricans, purification tablets (Aquatabs), and a container cover): 2,000 × €85 = €170,000, of 7,113 poor households.*

* Suggested formulation: check with a health worker before sharing.

Corrections: the draft said to log chlorine at each tanker fill and label failing tankers non-potable. On Réunion, tankers are deployed as non-potable ("non alimentaire") from the start, so the fix is the opposite: say where the drinking water is. The 3-of-6 under-chlorinated tankers (ARS, February 2025) were in the East, not CASUD. No current cut schedule for Saint-Joseph was found; *crise* is a restriction level. The panel's kits for all 7,113 poor households (€607,410) were cut to what one CCAS can hand out before the peak.

6. By 25 November, CASUD adopts its intercommunal safeguard plan with this annex, and it names a closure officer and physical barriers for the Saint-Joseph fords at orange heavy-rain vigilance

PARTIALLY CONFIRMED | PHASE: AVANT | HISTORIAN · EMERGENCY MANAGER · SKEPTIC | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
CASUD (president, by arrêté), with the four mayors and the préfecture	**25 November 2026 (legal deadline 26 November)**	**Already budgeted (drafting support €50,000–80,000)**	**A first cyclone season with no intercommunal plan; deaths at flooded crossings**

CASUD tendered for help to draft its PICS in 2025, with "épidémie" among its risks, and no adoption has been found. The commune plans it rests on are uneven: Le Tampon's dates from 2008, Saint-Joseph's from 2025, Entre-Deux's from 2023, and none is listed for Saint-Philippe. The annex adds what the commune plans miss: the water, heat, and epidemic triggers in this briefing, the rough-sleeper list (action 1), the Saint-Philippe stock (action 3), and the Saint-Joseph fords. On 28 August 2021, during floods on the Langevin and the Remparts, a person was swept away at Marine de Vincendo, Saint-Joseph, and the RN2 was cut. A closure notice at a ford is not a barrier: the plan names who puts a physical barrier across each *radier* at orange heavy-rain vigilance.

* Suggested formulation: check with a health worker before sharing.

Corrections: the draft said the Rivière des Remparts was unprotected until about 2030. The flood works were handed over on 13 December 2022 (about €17.9M plus €2.8M) and protect 290 people, 198 buildings, and the RN2; what remains open is the legal regularisation of the dyke system and the reinforcement of the right-bank cliff. The 2021 death is not recorded as being at a *radier*.

7. By 15 December, CASUD names the pumping stations and boreholes its €2 million generator line will cover and installs the first ones; it also asks Sudéau for a leak-repair deadline and brings network renewal for Saint-Joseph forward under FEDER

PARTIALLY CONFIRMED | PHASE: AVANT | ECONOMIST · EMERGENCY MANAGER · SKEPTIC | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
CASUD (water directorate and council), with Sudéau and the Région (FEDER managing authority)	Generator sites by 15 December 2026; contract and FEDER file in 2027	€2M in CASUD's 2026 budget (generators); renewal 75% FEDER	Power cuts that become water cuts after a cyclone; the slow loss of scarce water

CASUD's 2026 budget orientation report sets aside €3.5M for generators (€2M in 2026, €1.5M in 2027) in a single table row, with no sites named. Garance showed why it matters: the south was spared the worst wind but not the water and power cuts after it. The leaks are the slower problem. At 57.9% efficiency and 0.43% of pipes renewed a year, CASUD's network loses 22.4 m³ a day per kilometre (ILP, RPQS 2024). Its €15M efficiency budget puts €8.5M after 2028, beyond the 2021–27 FEDER programme, whose measure 2.5.2 pays 75% of pipe renewal (up to €500 a metre) on condition that efficiency improves. Bringing part of that forward for Saint-Joseph's sectors, and adding a leak-repair deadline to the Sudéau contract, both cost CASUD less than they save.

* Suggested formulation: check with a health worker before sharing.

Corrections: only the generators can add water before the peak; renewal and the contract change pay off over years, so they rank here, not higher. A contract change needs a negotiated amendment (*avenant*). FEDER 2.5.2 requires a water master plan less than five years old, and CASUD's is still being finalised, which may block a 2027 file. The "Chèque Eau" scheme is too small (about €12,000 a year for Saint-Joseph) and pays only bills, so it cannot fund kits.

8. Hold 1,500 three-day no-cook food kits at the Le Tampon and Saint-Joseph CCAS, released at pré-alerte or alerte orange before a forecast cut (never at alerte rouge, which is lockdown), and watch the monthly fresh-food price index

CONFIRMED | PHASE: PENDANT | FOOD SUPPLY · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
CCAS of Le Tampon and Saint-Joseph (épicerie solidaire, Pôle social, rue Paul Demange), with the Banque alimentaire des Mascareignes	Stock by 15 December 2026; released on the trigger	€39,000 (F04)	Households with food they cannot cook, and the poorest priced out after a cyclone

The food signal here is weak now and sharp after a cyclone. In August 2026, fresh-food prices were 2.2% lower than a year earlier (INSEE flash n°316). After Garance, they rose 19.1% in March 2025 and 7.1% more in April, and took until August to come back. The south grows much of the island's vegetables, and its growers wait: the State's fund approved compensation for Garance's farm losses fourteen months later (about €4.7M against more than €160M of losses). A food kit covers three days for a household at €26, on the Banque alimentaire's €3.70 a person a day (December 2022 study). The kits go out with the water kits, at pré-alerte or alerte orange, before a forecast or announced cut: at alerte rouge everyone is confined, and from a 72-hour cut the State already hands out bottled water. Trigger for scaling up food aid: fresh food up 10% or more on the month in the INSEE Réunion index, which is published about four weeks late.

* Suggested formulation: check with a health worker before sharing.

Note: these are the same CCAS and households as the island trigger cash (RE-02, €475 a household); release the kits first and the cash within seven days, from one list.

9. From November, fit covers on stored water and clear yard breeding sites door to door in Le Tampon and Entre-Deux, starting in Le Tampon's four priority neighbourhoods

PARTIALLY CONFIRMED | PHASE: AVANT | PUBLIC HEALTH · FRONTLINE · PHILANTHROPIST | ADAPTATION INVENTORY: NONE FOUND

OWNER	BY WHEN	COST	PREVENTS
The communes of Le Tampon and Entre-Deux, with neighbourhood associations under the ARS partner guide and the CCAS; the ARS is consulted	Rounds 2 November – 15 December 2026	€130,000: covers for 10,000 households at €13 delivered (F05)	Dengue bred in water stored during cuts

Le Tampon is at drought *alerte*, and its 9 October cut showed how fast households start storing water. Each visit removes breeding sites in the yard, fits one cover (a lid or mosquito mesh) to stored water, and repeats the island line RE-01. The 10,000 households sit inside the 14,111 poor households of Le Tampon and Entre-Deux; Saint-Joseph's and Saint-Philippe's poorest get their covers in the water kits (actions 3 and 5). The best trial is Camino Verde (Andersson and others, *BMJ* 2015): community mobilisation against breeding sites cut dengue infection in children by 29.5% (3.8–55.3%).

Corrections: ranked last because dengue is sporadic now: 1 to 5 cases a week in September 2026, with no commune named. The south was the epicentre in 2019, when dengue killed 14 people on the island, 9 of them directly (18,191 confirmed cases). Camino Verde studied *Ae. aegypti* in Nicaragua and Mexico, and covers were one choice within a wider campaign; a systematic review found the evidence for covers alone very thin (Bowman and others, 2016). The panel's cost comparator, Taborda and others 2022 (Girardot, Colombia), is confirmed: US$20.9 a household (PMC9294672), also for *Ae. aegypti*.

* Suggested formulation: check with a health worker before sharing.

WHO DOES WHAT

RACI CHART

R = Responsible (does the work) · **A** = Accountable (one per action; signs off and answers for it) · **C** = Consulted (asked before) · **I** = Informed (told after). Built from the owners named in the verified actions. No one here has been contacted or has agreed to a role; every row is a proposal. "Communes" are Le Tampon, Saint-Joseph, Saint-Philippe, and Entre-Deux.

Action	ARS	CASUD	Sudéau	Communes	CCAS	Croix-Rouge	Associations	Préfecture	Care providers
1. Rough-sleeper list and rounds	I	I		C	A/R	R	R	C	
2. Leptospirosis letter and vaccine	A/R	R	R	R			I		R
3. Saint-Philippe seven-day stock	I	C	C	A	R	R	R	I	
4. Dialysis plans and registers	A			C	R			I	R
5. Saint-Joseph notices and kits	C	A	R	C	R	R		I	
6. Adopt the PICS; ford barriers	C	A/R	C	R	C			C	
7. Generators, leaks, FEDER		A/R	R	I				I	
8. Food kits and price watch				I	A/R		R	C	
9. Covers door to door	C	I	I	A	R		R		

What the chart shows: the CCAS do the work in six rows and are accountable for the two that carry outreach and food (1 and 8). CASUD is accountable for the three water and planning rows (5, 6, and 7), and two of them need a decision rather than donor money: adopting the PICS and placing the generators. The ARS is accountable for two health rows (2 and 4). The communes are accountable for the cover rounds (9), because the ARS's own work is already State-funded, and Saint-Philippe's commune for its own stock (3), because it is the one that will be cut off. No row depends on the State paying first.

WHERE MONEY DOES THE MOST

FUNDING

The Philanthropist lens, reconciled against the ranked actions and verified. Placements are ranked by marginal lives saved per dollar. The envelope is bounded by what this place can spend before the peak, not by what it needs.

FLOOR

€295,197 (about $330,500)

TARGET

€496,651 (about $556,000)

CEILING

€904,028 (about $1,012,100)

LARGEST UNCERTAINTY

Whether the Saint-Philippe CCAS can buy, store, and release €135,000 of kits before December: no precedent was found

At the ECB reference rate of 14 October 2026, €1 = US$1.1195, fixed for all six Réunion briefings. Set unit costs: food kit €26, water-storage kit €85 (it includes a cover), cover €13 delivered. **Floor** = outreach €22,800 + Saint-Philippe water and food kits for its 927 poor households (927 ×

* Suggested formulation: check with a health worker before sharing.

€111 = €102,897) + 1,000 Saint-Joseph water kits (€85,000) + 750 food kits (€19,500) + covers for 5,000 households (€65,000) = €295,197. **Target** = outreach €22,800 + Saint-Philippe in full (927 water kits €78,795 + 2,156 food kits €56,056 = €134,851) + 2,000 Saint-Joseph water kits (€170,000) + 1,500 food kits (€39,000) + covers for 10,000 households (€130,000) = €496,651. **Ceiling** = outreach €22,800 + water and food kits for every Saint-Philippe household (2,156 × €111 = €239,316) + 4,000 Saint-Joseph water kits (€340,000) + 3,000 food kits (€78,000) + covers for the remaining 17,224 poor households (22,151 less 927 in Saint-Philippe and 4,000 with kits in Saint-Joseph; €223,912) = €904,028. Above that, one CCAS per commune binds before the peak, not money. **Not counted here:** the island trigger cash RE-F01 (€475 a household, in `reunion-island-funding.json`); CASUD's own generator line (€2M in 2026) and network renewal (action 7); the leptospirosis vaccine (ARS and CGSS); and the State's shelter and maraude budgets. **The target fell 38%** from the panel's €802,261: Saint-Joseph's kits went from all 7,113 poor households to 2,000, the outreach line from €60,000 to €22,800, and the covers ceiling no longer double-counts households that already get a cover in a kit.

RECIPIENT	USE	AMOUNT (TARGET)	MECHANISM	BASIS	VERDICT
CCAS of Le Tampon and Saint-Joseph, with the Croix-Rouge (Le Tampon antenne) and the Saint-Joseph Boutique Solidarité	Action 1: the named list, shelter kits, transport	€22,800 ($25,525)	Direct grant; readiness	300 shelter kits × €26 (€7,800) + transport to shelter €10,000 + list costs €5,000 (estimates). The cyclone deaths on record in CASUD, Belal (2024) and Gamède (2007), were people on the road or without shelter. No published cost-per-life figure	PARTIALLY CONFIRMED
Commune and CCAS of Saint-Philippe, with the Croix-Rouge / PIROI and the Banque alimentaire	Action 3: seven-day stock	€134,851 ($150,966)	Pre-positioned stock; released on the trigger	927 water kits × €85 + 2,156 food kits × €26. Ask PIROI what it already holds first. No cost-per-life figure for stock; the Bangladesh study is cash	PARTIALLY CONFIRMED
CCAS of Saint-Joseph, with the Croix-Rouge / PIROI	Action 5: water-storage kits for the poorest households	€170,000 ($190,315)	Pre-positioned stock; readiness	2,000 × €85, of 7,113 poor households (INSEE rate × households, approximation). No CASUD or State household-kit line exists for 2026	PARTIALLY CONFIRMED

* Suggested formulation: check with a health worker before sharing.

RECIPIENT	USE	AMOUNT (TARGET)	MECHANISM	BASIS	VERDICT
CCAS of Le Tampon and Saint-Joseph (épicerie solidaire), with the Banque alimentaire des Mascareignes	Action 8: food kits	€39,000 ($43,661)	Pre-positioned stock; released on the trigger	1,500 × €26 (€3.70 a person a day, BAM, December 2022, × household size × 3 days)	CONFIRMED
Neighbourhood associations under the ARS partner guide, with the CCAS of Le Tampon and Entre-Deux	Action 9: covers door to door	€130,000 ($145,535)	Pre-positioned stock; readiness	10,000 households × €13 delivered. Girardot, Colombia: a covers-and-curtains package at US$21.86 a house (Cortés and others, 2016), of which curtains were US$16.97	PARTIALLY CONFIRMED

The funder's one thing: CASUD will spend €2M on generators this year and is deferring €8.5M of leak repair past the end of the FEDER programme that would pay three-quarters of it. Donor money cannot fix that, and should not try. What nobody pays for is the last metre: a name on a list, a jerrican in a house in Saint-Philippe, a notice that says where the clean water is. That is where a euro placed in October still saves a life in February.

* Suggested formulation: check with a health worker before sharing.

Do not do this

CONTRAINDICATED · ADVICE THAT WOULD COST LIVES HERE

Advice that circulates for this hazard and would cost lives here. Cards 1–5 apply to every Réunion zone and are written once for the island (RE-X01 to RE-X05); cards 6–8 are specific to CASUD.

1. Do not call this season an "El Niño drought"

CONTRAINDICATED ISLAND-WIDE (RE-X01)

Météo-France expects El Niño to bring Réunion rain that is on average in surplus. CASUD's shortage is a carried-over deficit and networks that lose more than 40% of their water. A drought message tells people the rain will fix it, and it will not fix the leaks; and when the rain comes, it brings leptospirosis and floods at the fords.

Instead: use the premise: a record carried-over deficit and leaking networks, in a hot El Niño year.

2. Do not make insecticide fogging the visible response to dengue or chikungunya

CONTRAINDICATED ISLAND-WIDE (RE-X02)

Deltamethrin has been used on Réunion since 2006, and resistance in *Ae. albopictus* was detected in 2017. The ARS already limits spraying to the area around cases "to avoid creating resistance". Fogging looks like action and leaves the breeding sites in place.

Instead: keep spraying to the ARS's case-centred use, and put the effort into covered water (RE-01, actions 5 and 9).

3. Do not tell households to "store water" without saying "covered, emptied and scrubbed weekly"

CONTRAINDICATED ISLAND-WIDE (RE-X03)

Each cut or restriction notice makes households fill tanks, buckets, and drums: the containers mosquitoes breed in, just as the hot, wet season starts. All four CASUD communes are under drought restrictions now, and Saint-Joseph is at *crise*.

* Suggested formulation: check with a health worker before sharing.

Instead: "Couvrez vos réserves (couvercle ou moustiquaire), videz-les et frottez-les chaque semaine", on every water-cut notice (RE-01).

4. Do not push chikungunya vaccination with Ixchiq for people aged 65 and over, or put chikungunya vaccine stock ahead of dengue and leptospirosis

`CONTRAINDICATED` `ISLAND-WIDE (RE-X04)`

The HAS advised suspending Ixchiq for people aged 65 and over on 25 April 2025, after serious adverse events in older people, most of them on Réunion; that advice stands, and the EU restricted it further in 2026. The HAS estimates 66% of Réunion now carries chikungunya antibodies and judges a new epidemic "peu probable".

Instead: follow the HAS (24 April 2026): Vimkunya for people aged 65 and over and for 12–64-year-olds with comorbidities, only if an epidemic resumes. In CASUD, spend the effort on leptospirosis (action 2).

5. Do not let tanker water be taken as drinking water

`CONTRAINDICATED` `ISLAND-WIDE (RE-X05)`

In the January 2025 drought, the ARS recorded distribution units on tankers marked "non alimentaire", serving 36,240 people, and found three of six tankers it checked under-chlorinated and turbid. In Le Tampon on 9 October 2026, the tap water itself was declared unfit to drink after a breakdown.

Instead: label every tanker "non potable" in French and Creole, and hand out drinking water at the same point (action 5).*

6. Do not answer Saint-Joseph's groundwater *crise* with emergency boreholes or harder pumping of coastal wells

`CONTRAINDICATED` `CASUD`

The BRGM records seawater contamination of the south-coast aquifer between Saint-Pierre and Saint-Joseph, newly found between 1988 and 2005, with tide-linked salinity at Manapany-les-Bains, in Saint-Joseph. It calls saline intrusion irreversible. Groundwater was about 18% of Saint-Joseph's production in 2020. A well pumped harder in a drought can be lost for good.

* Suggested formulation: check with a health worker before sharing.

Instead: repair leaks first (Saint-Joseph's network delivered 50% in 2021; action 7), and put conductivity probes on the coastal wells, with a written chloride threshold at which pumping stops, as the BRGM recommends.

7. Do not issue a cut or restriction notice that does not say where to get safe water

CONTRAINDICATED CASUD

Le Tampon tied with Saint-Benoît for the most leptospirosis cases on the island this year, and Saint-Joseph had 15, in a record-dry season, and 19% of investigated cases had used untreated water for watering, irrigation, or washing. A notice that only lists cut hours leaves people to find water in ravines and basins. No source shows that restrictions themselves caused those cases: this card is a precaution drawn from the exposure data, not a measured effect.

Instead: every notice names the drinking-water point and carries the ravine-water warning (action 5).*

8. Do not plan to evacuate Saint-Philippe by road, or resupply it by tanker, at alerte rouge or while the Vincendo and Langevin crossings are in flood

CONTRAINDICATED CASUD

Saint-Philippe has one road. On 28 August 2021, floods cut the RN2 at Vincendo, isolated Grand Galet and Langevin, and swept a person away at Marine de Vincendo. At alerte rouge, all traffic is banned. A plan that moves people or water along the RN2 during the event depends on the road that fails first.

Instead: stock Saint-Philippe before December and shelter in place (action 3); move dialysis patients at pré-alerte (action 4).

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
1948 (26–27 Jan)	The "cyclone du siècle", island-wide (no ENSO record before 1950)	165 dead, island-wide; no commune breakdown found	—
1980 (Jan)	Cyclone Hyacinthe; record rain on the Remparts headwaters, 3,240 mm in 3 days at Commerson crater (neutral: ONI at +0.5 or more for only three seasons)	25 dead island-wide; no commune breakdown found	—
1989 (29 Jan)	Cyclone Firinga (La Niña, −1.6)	4 dead island-wide; deaths in the south not confirmed	—
2007 (Feb)	Cyclone Gamède (El Niño, +0.6)	2 dead on Réunion, one a driver from Le Tampon; about 90 injured; one of the two RN1 bridges at Saint-Étienne, the south's link to the west, collapsed; €143M of damage	Bridge rebuilt by 2013
2014 (2 Jan)	Cyclone Bejisa; gusts over 200 km/h at Le Tampon (neutral, −0.2)	1 dead island-wide	—
2019 (Jan–May)	Dengue epidemic, epicentre in the south: Saint-Joseph, Saint-Pierre, Le Tampon (El Niño, +1.0)	18,191 confirmed cases; 14 dead island-wide, 9 directly due to dengue	—
2021 (28 Aug)	Floods on the Langevin and the Remparts, Saint-Joseph (neutral, −0.5)	1 person swept away at Marine de Vincendo; RN2 cut; Grand Galet and Langevin isolated	Remparts flood works handed over in December 2022

* Suggested formulation: check with a health worker before sharing.

YEAR	EVENT	TOLL	WHAT CHANGED AFTER
2024 (15 Jan)	Cyclone Belal, the first violet alert (El Niño, +1.8)	3 homeless people dead island-wide; one in a tin shack in Le Tampon	The State's fund paid 56 uninsured people about €29,000, about six months later (11 July 2024)
2025 (28 Feb)	Cyclone Garance (neutral, −0.5)	5 dead island-wide, none in CASUD; water and power cuts in the south	EDF re-powered the Australine borehole at Saint-Philippe
2026 (13 Mar)	Lava crosses the RN2 at the Grand Brûlé, first time since 2007 (neutral, FMA +0.11)	No deaths; Saint-Philippe cut off from the east	Road fully reopened at the end of May
2026 (9 Oct)	Breakdown at the Leveneur plant, Le Tampon, during drought *alerte* (El Niño, +2.2)	No deaths; three days without safe tap water from the 14th km to the centre; do-not-drink order	Water packs handed out by the commune

The recurrence nobody has noticed: the cyclone deaths on record in CASUD, Belal (2024) and Gamède (2007), were people on the road or without shelter, and both storms came in El Niño years. The deadliest recent events were epidemics (dengue in 2019, with the south at its centre), not wind. And the water fails without a cyclone at all: the October 2026 breakdown at Le Tampon came before the season began. No death record by commune exists for CASUD, which is why most rows show island-wide tolls; the PICS should start one.

WHERE THE EXPERTS DISAGREE — UNRESOLVED

- **Put the money into household water kits** (philanthropist: one for every poor household in Saint-Joseph, €607,410) versus **put it into the network** (economist: a kit treats the symptom while half the water leaks). This briefing splits them by time: kits for 2,000 households, because only they help before the peak, and leak repair through FEDER and CASUD's own budget (action 7), because only it helps after.
- **Leptospirosis outranks dengue here** (public health and skeptic lenses, on this year's counts) versus **covers first** (the island premise, and dengue's 2019 epicentre in the south). This year's data favour leptospirosis; a dengue resurgence in the south would reverse the order of actions 2 and 9.
- **Saint-Philippe is cut off by lava** (several lenses) versus **lava cuts only the east side** (verifier). The verifier is right: tankers and food from Saint-Joseph still reach Saint-Philippe in an eruption. The stock is for a flood cut to the west or an alerte rouge.

* Suggested formulation: check with a health worker before sharing.

- **A salaried outreach post** (philanthropist) versus **the State already pays for maraudes** (verifier). Donor money funds the list, kits, and transport; if the State will not extend the Samu social to the Hauts, that is the ask.

WHAT NOBODY COVERED — BLIND SPOT

No lens found out how many people in CASUD live off the public network or on private catchments in the Hauts of Le Tampon and Saint-Joseph, or which of Sudéau's distribution units ran dry in 2025–26. The ARS holds the unit-by-unit list of disrupted distribution units, and the set has asked for it (data request no. 5). Until it arrives, actions 5 and 9 are targeted by commune and poverty, not by where the water actually stops. Nor did any lens count the dialysis patients in Saint-Philippe, which sets the size of the transport plan in action 4. And no lens checked whether the October 2026 Le Tampon outage was linked to a Sudéau strike, as one outlet suggested; if it was, the PICS needs a minimum-service clause, not a generator.

HOW TO READ THIS — METHOD

The panel is author-built. Eight simulated expert lenses were used to generate these recommendations. Where they agree, that is a strong hypothesis, not a consensus of the field. Every recommendation was then independently checked for evidence, transferability to this location, and feasibility before the peak; the verdict tags carry the result.

Mortality figures are resolved at Climate Impact Lab impact-region scale. They are not neighbourhood figures. The El Niño projection covers *heat only*; drought, flood, fire, smoke, and crop failure are additional to it and are carried here by the historical record rather than by a published number. The study's map publishes rows for Réunion's four arrondissements. CASUD sits in the Saint-Pierre row (2), which it shares with CIVIS and Cilaos (318,130 people in all). Scaled by population, CASUD's share is 2 × 134,044 / 318,130 = 0.84. That is our arithmetic, not an Impact Lab finding, and it ignores that the Hauts of Le Tampon are cooler than the coast.

HOW THIS BRIEFING WAS PRODUCED — METHODOLOGY

- **Scope.** The location was taken as given: CASUD (Le Tampon, Saint-Joseph, Saint-Philippe, Entre-Deux), the fourth zone of six in the Réunion set. Climate Impact Lab impact region, as the report records it: REU.Re7bdb3a99bec931b, the whole island (about 890,000 people; label unconfirmed). CASUD has no row of its own: it is part of the Saint-Pierre arrondissement row of the study's map. Commune detail is used for hazard exposure and action design, not for death counts.

* Suggested formulation: check with a health worker before sharing.

- **Exposure.** Climate Impact Lab, *451,000 at Risk* (September 2026, pre-peer-review): ECMWF SEAS5 seasonal forecasts (51-member ensemble) applied to temperature–mortality relationships across 24,378 regions, against a 1996–2025 baseline, for June 2026 – February 2027. It covers heat only. The arrondissement rows are the 6-month field (`6mo_hotonly_total_mean`) of the study's map (Flourish visualisation 30298258), as resolved for the CINOR briefing; the CASUD figure is a population scaling of the Saint-Pierre row, shown with its arithmetic. It is never added to the Impact Map's long-run projections or quoted from the map's GDP-denominated mortality. Water, leptospirosis, isolation, and cyclones are carried by the historical record.
- **Expert panel.** Eight simulated expert lenses (emergency manager, public health scientist, frontline responder, skeptic, economist, historian, philanthropist, and food supply chain analyst) each researched this place independently and in parallel, from web sources, to a fixed brief. Each worked from a shared pack of facts already verified for the Réunion set, and each was told to re-check any fact it used.
- **Reconciliation.** Duplicate actions were merged and backing lenses recorded. Convergence was tested for shared bias (whether the evidence behind an agreed action came from a setting like this one) before it counted in an action's favour. Contradictions were kept, not resolved by dropping a side. Actions were ranked by expected lives saved per unit of effort before the peak, each was given a named payer and trigger or marked unfunded, and the funding placements were reconciled against the ranked list.
- **Verification.** Five verifier agents checked every recommendation on three tests: evidence (the primary source and its effect size), transferability (whether that evidence came from somewhere like here), and feasibility (whether it can be done here before the peak). The water and finance lines were also checked for a named payer, a trigger, and whether they already exist. The funding envelope was checked for real recipients, published unit costs, and duplication, then re-summed. The Do not cards, the scope table, and every row of the record were checked at source, with the NOAA ONI for each season.
- **Who does what.** The RACI chart is built from the owners named in the verified actions, with one accountable party per action. It is a proposal: no organisation named in it has been contacted or has agreed to a role.
- **Verdicts.** *Confirmed*: evidence, transferability, and feasibility all hold. *Partially confirmed*: holds after the corrections shown. *Unverified*: could not be checked to a primary source. *Contraindicated*: would do harm or waste the window here.
- **Adaptation Inventory.** Consulted for the set (the 96 entries embedded in the inventory page, read for the CINOR briefing). It holds no entry for France, Réunion, small islands, leptospirosis, dialysis, or water rationing. The only near precedent, anticipatory cash for floods (Bangladesh, Nepal), applies to the island cash line.
- **Health wording.** Passages where a wording error could hurt someone (leptospirosis, its vaccine, and when to see a doctor, ravine water, and drinking water) are marked "* Suggested formulation: check with a health worker before sharing." They were checked against published

* Suggested formulation: check with a health worker before sharing.

sources, not by a health professional. They are suggestions, not medical advice, and anyone passing them on should confirm them with a doctor, pharmacist, nurse, or community health worker first. The marker was added on 2 November 2026.

- **Rules.** No cost is derived by dividing a published total, and no local figure is invented; where data is thin, the report says so and labels its estimates. One exchange rate (ECB, 14 October 2026) and one set of unit costs are used across the Réunion set. Every source is listed below with its verification status.
- **Limits.** The panel and verifiers are AI research agents working from web-available sources, not human experts, and no one in this place was interviewed. The web-search cap was not reached in this run. Légifrance and several préfecture and DEAL pages returned 403, so the QPV list was checked against INSEE's table, not the decree. Commune-level death tolls before 2024 could not be found, apart from Gamède's Le Tampon driver (2007); most record rows give island-wide tolls. Current cut schedules for Saint-Joseph and Saint-Philippe were not found. Two inherited facts were corrected in this run (the South's leptospirosis rate, and the Remparts works). The deficit wording was checked against Météo-France's own provisional season review (29 April 2026): the driest rainy season in 54 years of records.
- **Revision.** Revised 2026-10-15 after the set audit (fixes applied: Gamède, 2 dead on Réunion, one from Le Tampon, and one of the two RN1 bridges; the 2019 dengue toll, 18,191 cases and 14 dead, 9 direct; the Météo-France deficit wording and cyclone clause; the food-kit release moved to pré-alerte or alerte orange, in line with the set; leptospirosis communes, with Saint-Benoît tied at 20; the vaccine schedule and timing; the Belal shelter-refusal wording and the six-month relief payment; the register covers people aged 65 and over; the communes are accountable for the cover rounds; Saint-Philippe kits released at pré-alerte or alerte orange; the linear loss index, 22.4 m³ a day per kilometre; the RN2 detour restored; Taborda confirmed; Hyacinthe and the March 2026 lava cut labelled neutral; the impact-region size clause removed; funding records store the ceiling; island-wide events linked to the national file).

References · status after verification in brackets

Exposure: Climate Impact Lab, 451,000 at Risk and its map sep-feb-heat-deaths [arrondissement row confirmed for the set; CASUD figure is our scaling] · Impact Map boundary file [region decoded; label unconfirmed] · Adaptation Inventory [consulted]

Place and water: INSEE dossiers complets, Le Tampon, Saint-Joseph, Saint-Philippe, Entre-Deux [confirmed] · QPV list 2024 [confirmed; decree not checked] · VigiEau, 14 Oct 2026 [confirmed] · Prefectoral restriction order, 9 Oct 2026 [confirmed] · Office de l'eau, Chroniques de l'eau 133 [commune efficiency confirmed] · CASUD, RPQS drinking water 2024 [confirmed] · CASUD, ROB 2026 [confirmed] · CASUD, Chèque Eau 2026 [confirmed] · Région, FEDER fiche 2.5.2 [confirmed] · Imaz Press, Le Tampon cut, Oct 2026 and Réunion La 1ère [confirmed] · ARS drought sheet, 6 Feb 2025 [confirmed; tankers in the East] · BRGM/RP-59457-FR, coastal aquifers [confirmed] · Préfecture, Cazala intake inquiry (Saint-Joseph supply mix) [confirmed] · Météo-France, 2025–26 rainy-season review (provisional), 29 Apr 2026 [confirmed for the set]

Health: SPF leptospirosis bulletin, 17 April 2026 [corrected: 39% of cases, not 39 per 100,000] · Lebeau-Desmoulin et al., treatment delay [confirmed] · ARS leptospirosis vaccine leaflet [confirmed] · SPF Réunion bulletin, 9 Oct 2026 [confirmed] · Préfecture and ARS, dengue 2019

* Suggested formulation: check with a health worker before sharing.

review, 7 Jan 2020 [confirmed: 18,191 cases, 14 deaths, 9 direct] · Senate report r25-554 [confirmed] · Lurie et al., AJKD 2015 [confirmed; observational] · ARS, dialysis units [confirmed] · Andersson et al., BMJ 2015 (Camino Verde) [confirmed; other vector] · Bowman et al., 2016 [confirmed] · Cortés et al., Girardot costs [confirmed] · Taborda et al. 2022 [confirmed] · HAS, Ixchiq, 25 Apr 2025 [confirmed]

The record and access: Belal deaths, Outremers360 (AFP) and Imaz Press [confirmed: one Le Tampon death] · Fondation Abbé Pierre, Réunion 2024 [confirmed] · Croix-Rouge Le Tampon antenne [confirmed] · OVPF, 24 Mar 2026 [confirmed] · Imaz Press, RN2 reopening [confirmed] · RCI, RN2 detour, 13 Mar 2026 [confirmed: 30 minutes to 2.5 hours] · La 1ère, Saint-Joseph floods, Aug 2021 and the victim [confirmed] · DEAL, Remparts works [confirmed: handed over Dec 2022] · CASUD, PICS drafting [confirmed: in preparation] · EDF, Garance update [confirmed] · Cyclone Hyacinthe · Cyclone Firinga [confirmed, island-wide tolls] · CGEDD, Gamède and the Saint-Étienne bridge [confirmed via lens; not re-checked] · Cyclone Gamède [set value: 2 dead on Réunion, one from Le Tampon] · Cyclone Bejisa [island-wide toll; secondary] · NOAA CPC, ONI [confirmed]

Food and money: INSEE flash n°316, Aug 2026 prices and after Garance [confirmed] · Farm-loss compensation, Apr 2026 [confirmed] · BAM €3.70 a day [confirmed; 2022 figure] · Saint-Joseph épicerie solidaire [confirmed] · Pople et al., WFP 2021 [confirmed; cash, not stock] · Crédit Agricole after Garance [confirmed; post-event]

Exposure data · Climate Impact Lab, 451,000 at Risk · Impact Map · Adaptation Inventory

Generated 14 October 2026 for South (CASUD), Réunion. El Niño Ready is volunteer-run and is not affiliated with Climate Week NYC, The Climate Group, or Columbia University.

* Suggested formulation: check with a health worker before sharing.